

Nilanjan Debnath

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PROFESSIONAL EXPERIENCE

Locus, Bangalore, India

Sept 2025 – Present

Data Scientist-2

- Integrated and developed clustering into Locus' last mile routing engine which enhanced compactness and increased allocations for clients across continents.
- Built a delivery linked checkout platform for IKEA using XGBoost which brought down processing time to milliseconds while retaining 99.9% of accuracy
- Maintained routing engine reliability by fixing critical bugs and managing end-to-end production deployments and versioning cycles.

The University of Texas at Dallas, Dallas, TX

Aug 2022 – Dec 2024

Eugene McDermott Fellow Graduate Research Assistant

- Estimated the upper and lower bounds for school admissions in centralized systems using two-sided matching strategies potentially decreasing wasteful applications by 7.9%.
- Analyzed, identified and presented upon key variables affecting match quality in foster care adoptions, resulting in a potential 12% reduction in the rate of children who reenter foster care.

Mercedes Benz Research and Development, Bangalore, India

Feb 2022– Jul 2022

Computer-Aided Engineering Analyst

- Enhanced ergonomic satisfaction of Mercedes carlines by 12% through in-depth analysis of Mercedes Benz and benchmarking competitor cars using advanced computer-aided design (CAD) tools.
- Automated ergonomic analysis processes with computer vision, reducing analysis time by 82%.
- Established the process for properly simulating accident scenarios using Generative Adversarial Networks (GANs) to generate 25% additional accident data which was useful for accident prevention scenarios.
- Led reinforcement learning workshops to accelerate project development and enhance team proficiency.

DeLab Research, Trivandrum, India

Jan 2021 – Apr 2021

Machine Learning Intern

- Optimized next-day renewable energy forecasting by deploying a host of regression models on Spark based ETC pipelines in Azure Databricks to achieve 90% prediction accuracy.
- Surpassed industry benchmarks by enhancing operational efficiency and cutting forecasting errors by 8%.

Coriolis Technologies, Pune, India

May 2020 – July 2020

Machine Learning Intern

- Fortified information security by utilizing Named Entity Recognition techniques with NLP frameworks (SpaCy & Polyglot) to conceal named entities to achieve a 94% true positive classification rate.
- Applied deep neural networks on OCR-extracted text for high-precision document classification and contributed significantly to the GitHub repository of Microsoft Presidio.

Indian Statistical Institute

Research Intern

May 2021 – Dec 2021

- Engineered clustering algorithms for streaming data using decision-theoretic rough sets.

Indian Statistical Institute

Research Intern

Dec 2019

- Assisted in data pre-processing and data classification based on Bayesian decision theory.

EDUCATION

The University of Texas at Dallas

May 2025

Master of Science, Supply Chain Management (Eugene McDermott Fellow, Transferred from PhD in OR)

Chennai Mathematical Institute, Chennai

May 2021

Master of Science, Data Science

St. Xavier's College, Kolkata

May 2019

Bachelor of Science, Mathematics

ADDITIONAL INFORMATION

Tools: SQL, Python, R, Power BI, MATLAB, Tableau, Microsoft Excel, AWS, Databricks, Microsoft Powerpoint

Skills: Statistical methods, Linear Regression, Machine Learning (Supervised and Unsupervised), Natural Language

Processing, Deep Learning, Reinforcement Learning, Computer Vision, Generalized Linear Models, Regularization, Time

Series Analysis, Forecasting, Monte Carlo Simulation, Linear Programming, Root cause analysis, A/B testing